MAT 017B Quiz 8

September 11, 2018

Name: Key

SID:

1. Find the eigenvectors of the matrix

$$A = \begin{bmatrix} -2 & 4 \\ -3 & 5 \end{bmatrix}$$

$$\Delta = (-2)(5) - (-3)(4) = -10 + 12 = 2$$

$$\tau = -2 + 5 = 3$$

$$\lambda^2 - 3\lambda + 2 = 0$$

$$(\lambda - 2)(\lambda - 1) = 0$$

$$\lambda = 1, 2$$

$\underline{\lambda = 1}$:

$$\begin{bmatrix} -2-1 & 4 \\ -3 & 5-1 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 4 \\ -3 & 4 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$-3u_1 + 4u_2 = 0$$

$$3u_1 = 4u_2$$

$$\boxed{\vec{u} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}}$$

$\underline{\lambda = 2}$:

$$\begin{bmatrix} -2-2 & 4 \\ -3 & 5-2 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 4 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-4v_1 + 4v_2 = 0$$

$$v_1 = v_2$$

$$\boxed{\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$